

A Research Principle for the AI Era

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Even when intrinsic value is the sole objective, the optimal strategy begins by devoting all or nearly all resources to external value.

Two results establish this principle for a sufficiently long research horizon.

Intrinsic value is the scientific contribution the researcher ultimately values. External value is support convertible into research funding, measured in the same units as expenditure.

Goal

Maximize expected total intrinsic value over n research cycles:

$$\max \mathbb{E} \left[\sum_{i=1}^n I_i \right].$$

With linear expected returns, a long horizon makes full initial external focus optimal.

Assumptions for Theorem 1

Consider n research cycles, starting with a seed budget $c_1 = 1$. In cycle i , expenditure is c_i and the fraction allocated to intrinsic work is $p_i \in [0, 1]$. Write $\mathbb{E}_i$ for expectation conditional on the information available when p_i is chosen, including that choice.

1. Linear expected returns. For every decision history and allocation,

$$\mathbb{E}_i[I_i] = ap_i c_i, \quad \mathbb{E}_i[E_i] = w(1 - p_i)c_i, \quad c_{i+1} = d_i E_i.$$

The constants $a > 0$ and w are common to all cycles. The multipliers $d_i \geq 1$ are iid and bounded; each draw is independent of the decision information and current external return.

2. A growing reserve. For every decision history and allocation, the lowest possible external return is $b(1 - p_i)c_i$, with the same b across cycles. The funding balance obeys

$$B_0 = 0, \quad B_i = B_{i-1} + E_i - c_i.$$

For every possible outcome, require

$$B_i \geq \rho B_{i-1}, \quad 1 < \rho < b.$$

Research income must cover spending while growing the reserve.

3. Decision information. The researcher observes the current budget and balance and chooses p_i before the current returns and funding multiplier are drawn. Only b , w , and $\mathbb{E}[d_i]$ are known about their distributions.

Full initial external focus

Theorem 1. *There is a finite integer $K \geq 1$, independent of the horizon n , such that whenever $n > K$, the unique optimal decision in each of the first $n - K$ cycles is to allocate all resources to external value:*

$$p_i = 0 \quad (1 \leq i \leq n - K).$$

A numerical example

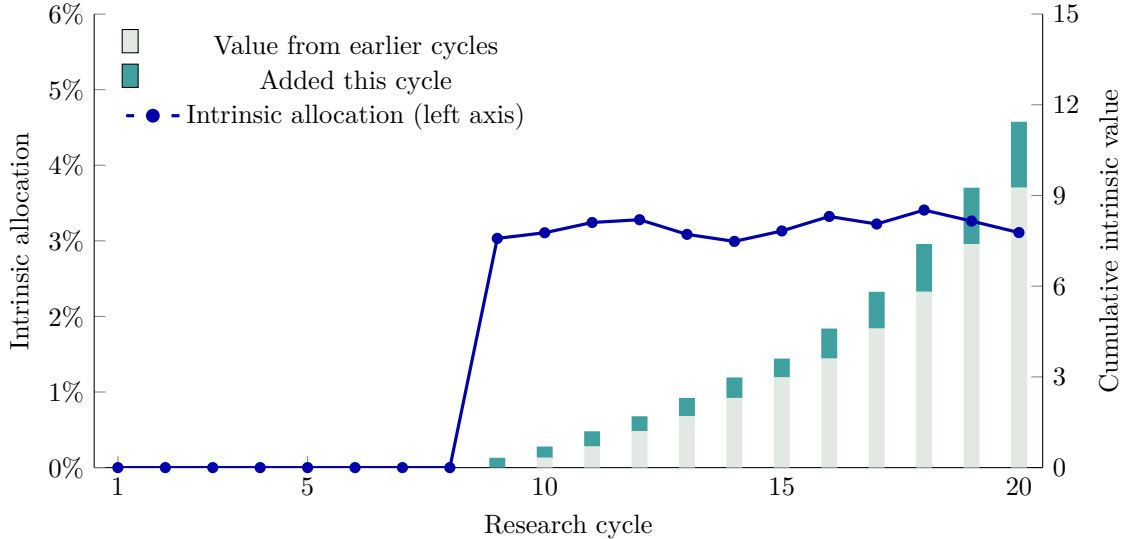


Figure 1: One optimal trajectory with iid random coefficients. All resources go to external value in the first eight cycles. The line shows the intrinsic allocation share. Each bar shows cumulative intrinsic value, with the current increment highlighted.

Use $I_i = a_i p_i c_i$ and $E_i = b_i (1 - p_i) c_i$, with $n = 20$, $\rho = 1.045$, and independent uniform draws

$$a_i \sim U[1.5, 2.5], \quad b_i \sim U[1.05, 1.15], \quad d_i \sim U[1, 1.2].$$

Here $K = 12$, so the first eight cycles are devoted entirely to external value. The figure shows one realization with random seed 20260928.

Proof of Theorem 1

Every feasible state satisfies $0 \leq B \leq c$: this holds initially and is preserved because

$$0 \leq B' = B + E - c \leq E \leq d_i E = c'.$$

Write $u = (1 - p)c$ for external expenditure. The reserve constraint gives the feasible interval

$$\frac{c + (\rho - 1)B}{b} \leq u \leq c.$$

Divide the objective by a , giving expected stage reward $c - u$. Independence of the funding multiplier gives

$$v = w\mathbb{E}[d_i] \geq w \geq b > \rho > 1, \quad \mathbb{E}_i[c_{i+1}] = v(1 - p_i)c_i.$$

At every decision history with state (c, B) and m cycles remaining, the value is $V_m(c, B) = A_m c + D_m B$, with $A_0 = D_0 = 0$. To see this, define $S_m = vA_m + wD_m$. The Bellman objective is

$$(1 - D_{m-1})c + D_{m-1}B + (S_{m-1} - 1)u.$$

Its maximizing endpoint depends only on S_{m-1} , and both endpoints are linear in (c, B) , completing the induction.

When $S_{m-1} \leq 1$, choose the lower endpoint. Starting from $S_0 = D_0 = 0$, this gives

$$D_m = D_{m-1} + \frac{(\rho - 1)(S_{m-1} - 1)}{b} \leq 0$$

and

$$\begin{aligned} S_m &= \frac{v + w(\rho - 1)}{b} S_{m-1} + \frac{v(b - 1) - w(\rho - 1)}{b} \\ &\quad + (w - v)D_{m-1} \geq \rho S_{m-1} + b - \rho. \end{aligned}$$

Since $\rho > 1$ and $b > \rho$, this sequence crosses 1 in finite time. Let $K = \min\{m \geq 1 : S_m > 1\}$.

Once $S_{m-1} > 1$, the unique maximizing choice is $u = c$. Then $D_m = D_{m-1} \leq 0$ and

$$S_m = vS_{m-1} + (w - v)D_{m-1} > S_{m-1} > 1.$$

Thus every decision with more than K cycles remaining uniquely allocates everything to external value. These are precisely the first $n - K$ cycles. $\square$

With nonlinear returns that retain a positive minimum gain from added investment, the optimal initial external share still approaches one.

Assumptions for Theorem 2

Keep the same objective, initial budget and balance, exact budget rule, and growing-reserve requirement. Use the following assumptions for nonlinear returns.

1. Random return functions. Let $F_i(x)$ and $G_i(x)$ be the intrinsic and external returns from expenditure x , so

$$I_i = F_i(p_i c_i), \quad E_i = G_i((1 - p_i)c_i), \quad c_{i+1} = d_i E_i.$$

The nonnegative return functions vanish at zero. The pairs (F_i, G_i) are iid across cycles. The multipliers $d_i \geq 1$ are iid, bounded, and independent of the return functions. Allocations are chosen before the current draws. At every feasible state, finite-horizon optimal policies exist and have finite expected value.

2. Positive return increments. Write $f(x) = \mathbb{E}[F_i(x)]$. For some $m > 0$ and every $x, h \geq 0$, require

$$f(x+h) - f(x) \geq mh, \quad G_i(x+h) - G_i(x) \geq bh.$$

The second inequality holds for every realization. At each expenditure x , the lowest possible external return is bx , with $1 < \rho < b$ as before. These conditions allow nonlinear returns with a positive minimum gain from additional investment.

Approximate initial external focus

Theorem 2. *For every horizon $n \geq 2$, the first intrinsic share $p_1^{*(n)}$ under any optimal policy satisfies*

$$0 \leq p_1^{*(n)} \leq \frac{f(1)}{m(b-\rho)b^{n-2}}.$$

Consequently, $p_1^{(n)} \rightarrow 0$ as $n \rightarrow \infty$: the optimal initial allocation becomes arbitrarily close to full external focus.*

A nonlinear example

Use $n = 100$, $\rho = 1.01$, and

$$I_i = p_i c_i + A_i \sqrt{p_i c_i}, \quad E_i = 1.05(1 - p_i)c_i, \quad d_i = 1,$$

with iid $A_i \sim U[0.8, 1.2]$. Then $f(x) = x + \sqrt{x}$, so the second set of assumptions holds with $m = 1$ and $b = 1.05$. We numerically maximize expected total intrinsic value and show one realization with random seed 20260928.

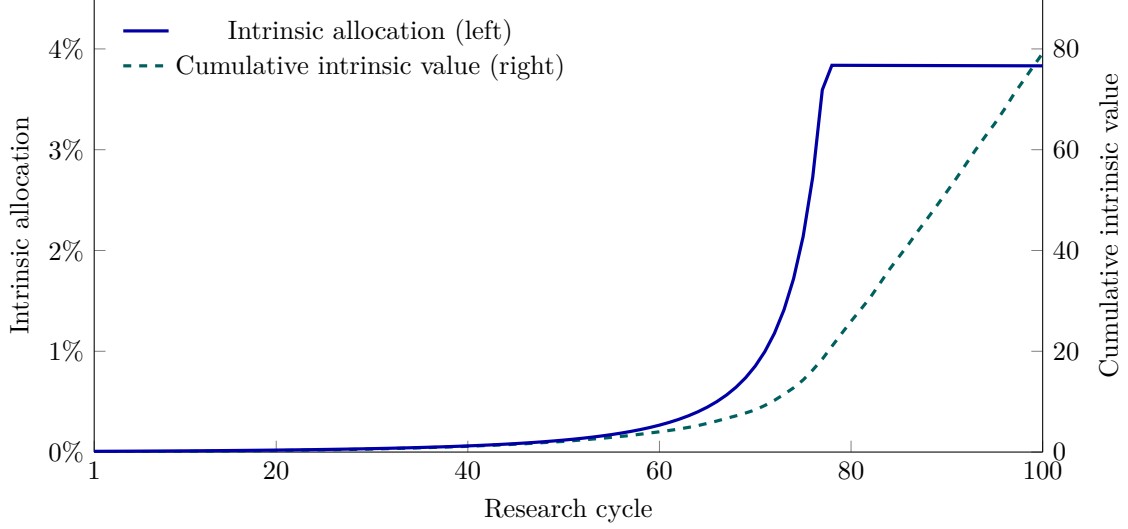


Figure 2: One 100-cycle realization of the nonlinear model. The solid line shows the intrinsic allocation share; the dashed line shows cumulative realized intrinsic value. The first share is 0.0074 percent, rising to about 3.8 percent in later cycles.

Proof of Theorem 2

As before, $0 \leq B \leq c$. Intrinsic expenditure t is feasible precisely when

$$0 \leq t \leq \frac{(b-1)c - (\rho-1)B}{b}.$$

Let $V_k(c, B)$ denote optimal expected intrinsic value with k cycles remaining. We first show that increasing the state by (δ, η) , with $0 \leq \eta \leq \delta$, gives

$$V_k(c + \delta, B + \eta) - V_k(c, B) \geq L_k \delta, \quad L_k = m(b - \rho)b^{k-2}.$$

For $k = 1$, optimal intrinsic expenditure is the upper endpoint above. It increases by at least $(1 - \rho/b)\delta$, so the lower bound on f gives $L_1\delta$.

For the induction step, use the smaller state's optimal intrinsic expenditure in the larger state. External expenditure increases by δ ; the reserve constraint remains satisfied because $b\delta \geq \delta + (\rho - 1)\eta$. Couple the two decisions using the same return functions and funding multiplier. The external return increases by some $\Delta E \geq b\delta$. The next-state differences are

$$\delta' = d_i \Delta E, \quad \eta' = \eta + \Delta E - \delta,$$

so $0 \leq \eta' \leq \delta'$ and $\delta' \geq b\delta$. Current intrinsic returns are identical. The induction hypothesis therefore gives a continuation gain of at least $bL_{k-1}\delta = L_k\delta$.

Finally, at the initial state $(1, 0)$, compare an optimal intrinsic expenditure $t = p_1^{*(n)}$ with full external allocation. Full external allocation increases the next budget and balance by $(d_i \Delta E, \Delta E)$, where $\Delta E \geq bt$. The value comparison just proved gives a continuation gain of at least $bL_{n-1}t = L_nt$. Optimality of t requires its current expected intrinsic return to cover this gain:

$$L_nt \leq f(t) \leq f(1).$$

Dividing by L_n proves the bound. Since $b > 1$, its right-hand side tends to zero. $\square$

Even when only intrinsic value matters, the optimal starting allocation is entirely or almost entirely external.